

3. To show that $L_{p,q}(X, Y)$ is closed in $(L(X, Y), \|\cdot\|)$ let (A_n) be a sequence in $L_{p,q}(X, Y)$ which is $\|\cdot\|^*$ -convergent to some $A \in L(X, Y)$. For $\epsilon = 1$ let $n_0 \in \mathbb{N}$ be such that $\|A_n - A\|^* \leq 1$ for all $n \geq n_0$. Then $\|A\|^* \leq \|A - A_{n_0}\|^* + \|A_{n_0}\|^* \leq 1 + \|A_{n_0}\|^*$, which implies that both A and $-A$ belong to $L_{p,q}(X, Y)$. $\square$

Remarks 3.8. 1. If a sequence (A_n) in $L_{p,q}(X, Y)$ converges to $A \in L(X, Y)$ with respect to the conjugate norm $(\|\cdot\|_{p,q}^*)^{-1}$ of $\|\cdot\|_{p,q}^*$, then $A \in L_{p,q}(X, Y)$.

2. As it is known, in the classical case, the completeness of $(L(X, Y), \|\cdot\|)$ implies the completeness of the normed space Y . We do not know if a similar result holds for the extended norm $\|\cdot\|^*$.

Let n_0 be such that $\|A - A_n\|^* \leq 1$ for all $n \geq n_0$. Then $\|A\|^* \leq \|A - A_{n_0}\|^* + \|A_{n_0}\|^* \leq 1 + \|A_{n_0}\|^* < \infty$, shows that $A \in L_{p,q}(X, Y)$, proving the validity of the assertion from 1.